



Research paper

Techno-economic and environmental estimation assessment of floating solar PV power generation on Akosombo dam reservoir in Ghana

Mohammed Okoe Alhassan^{a,b}, Richard Opoku^{b,c}, Felix Uba^{d,e,*}, George Y. Obeng^b, Charles K.K. Sekyere^b, Peter Nyanor^b

^a Department of Mechanical Engineering, Takoradi Technical University, P.O. Box 256, Takoradi, Ghana

^b Department of Mechanical Engineering, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

^c The Brew-Hammond Energy Center, College of Engineering, KNUST, Ghana

^d Department of Mechanical and Manufacturing Engineering, University of Energy and Natural Resources, Sunyani, Ghana

^e Regional Centre of Energy and Environmental Sustainability (RCEES), UENR, Ghana



ARTICLE INFO

Article history:

Received 9 May 2023

Received in revised form 15 August 2023

Accepted 11 September 2023

Available online 22 September 2023

Keywords:

Floating Solar Photovoltaic (FPV)

Evaporation

RETScreen Expert

Hydropower

Renewable Energy (RE)

ABSTRACT

Ghana has developed energy policies to help increase the use of renewable energy in its energy mix. With abundant water reservoirs and solar irradiation, the potential to deploy floating solar photovoltaic is feasible to increase the country's renewable electricity generation. RETScreen Expert software was used for studying a proposed FPV-Hydro hybrid plant system. This study conducted a feasibility analysis for a 420 MWp FPV on Akosombo Dam reservoir a location with 4.66 kWh/m²/day solar energy. The study recommended FPV power plant with capacity factor of 14.1%, and would consist of 500,000 units of solar panels covering a minimum area of 2,460,457 m² to generate a total annual electricity of 520,233 MWh. The project will save 73,327 cubic metres of water from evaporating which can produce approximately 10 MWp hydroelectric power annually. FPV economic analysis shown that it will results in lower LCOE of US\$ 0.10/kWh, annual revenue of USD \$52,238,576.00 and 12-years simple payback time, indicating positive economic indicators. Additionally, it will significantly reduce GHG emissions by 308,904.5 tCO₂/MWh annually. The FPV-Hydropower hybrid plants prove feasible and contributes to a greener and less costly energy generation system to meeting the 10% additional Renewable Energy (RE) target of Ghana.

© 2023 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Better living standards and economic development is globally dependent on access to energy because of their positive relation (Qi et al., 2021; Opoku et al., 2020b). Meeting the global energy demand must be sustainable as it is critical for industrialization and growth. For the past several years, energy demands at the global, regional and national levels have been increasing. For example, total electricity demand globally increased by 2.3%, reaching 26,100 TWh in 2018 (Philip et al., 2019). About 65.3% of the energy generated are from conventional fossil fuel sources, which are harmful to the environment leading to climate change issues (IEA, 2019; Global Energy Review 2020, 2020).

Global investments in low-carbon energy transformational options, such as renewable energy (RE) as strategic sustainability

measures, has resulted in a worldwide reduction of toxic emissions from fossil-fuel sources (Gunfaus and Waisman, 2021; Zhao et al., 2020; Opoku et al., 2018; Ahadzie et al., 2020). Global regions such as Ghana are exploring RE alternative energy sources, such as; bioenergy, hydro, solar and wind for electricity generation (Huld and Commission, 2014). Amongst the RE sources, solar PV electricity generation systems with appreciable solar irradiation levels (>13 MJm²/day) are proving as cost competitive compared with utility grid powered systems from fossil-fuelled systems, due to its energy market value for domestic and industry applications (Opoku et al., 2020a). For the past decade, prices of solar PV panels have dropped significantly, increasing installation rate for both residential and commercial sectors, which has promoted clean energy technologies and significantly helped to improve the adverse impact of global climatic variation (Rauf et al., 2019; Teixeira et al., 2015).

Most solar PV power plants in Sub-Saharan Africa are ground-mounted and consumes large areas for installation (Hove et al., 2020). Increasing concern of deforestation for solar PV installations coupled with land competition for crop production and

* Corresponding author at: Department of Mechanical and Manufacturing Engineering, University of Energy and Natural Resources, Sunyani, Ghana.
E-mail address: felix.uba@uenr.edu.gh (F. Uba).

estate developments, creating the need to develop and deploy new technologies such as floating solar PV (FPV) systems (Sahu et al., 2016). FPV shares several advantages over ground mounted PV installations. Floating PV tends to be about 11% more performance efficient than the land-based solar panel as a result of the temperature reduction effect in water (Monforti-ferrario et al., 2016; Ates et al., 2020). Performance of PV systems are reduced when panel thermal stress is high due to high temperature coefficient. In other words, the floating PV is advantageous in terms of land and water conservation, environmental aspect (algae formation), as well as less panel cleaning and maintenance cost due to absence of dust. Due to this benefits, FPV systems installed on water-bodies, such as lakes, reservoirs/dam and ponds, have significant already been deployed across regions and increased globally. Worldwide installed capacity for FPV has increased significantly with enormous technological growth since 2016 (Farfan and Breyer, 2018; Lee et al., 2020).

However, implementing this FPV technologies requires careful consideration of the available technical, economic and environmental impacts. Therefore, there is a need for a robust and effective feasibility and system performance approach that can prioritize its technological resources for sustainable electricity generation.

Presently, studies on the performance of installed floating solar PV (FPV) power generation technologies conducted to improve its utilization exist in literature. For instance, Durkovic and Djuricic (2017) investigated the environmental impact of a large-scale floating photovoltaic power system in Montenegro. Their study showed significant reduction in CO₂ emissions of about 83.42 ktons CO₂/year, water evaporation savings of about 5.41 million m³/year. In addition, the study showed that Montenegro PV plant's efficiency could be maximized by 16% (IRR = 16%) at recommended power plant consisting of 18 plants to generate a total 90 MWp installed power. Kim et al. (2019) explored the possibility of installation FPV in Korea to give 2932 GWh power. In addition, greenhouse gas (GHG CO₂) emission analysis showed an estimated annual reduction of 1,294,450 tons per year. Another study conducted by Rauf et al. (2020) explored the potential of hybridized 'hydropower-FSPV' for a small dam in Pakistan. These studies analysed the performance of a 500 kV, 132 kV and 11 kV voltage levels for the combined integration of hydroelectric power and FSPV plant, implementing and incorporating an optimization model to maximize power output and reach peak demand, to analyse the plants technical feasibility. The results proved an additional 3.5% power output for the combined operation for the combined system consisting of hydroelectric and 200 MWp FSPV. However, the study does not provide any economic or environmental views. The study concluded that, the project could provide additional electricity and reduce the need for new hydroelectric generators. Other studies revealed that power plants with FPV capabilities could prevent up to 74 billion m³ of water loss, increasing water accessibility estimated at 6.3 per cent, which would further benefit reservoir-based hydropower production from FPV plants, by an estimated 142.5 TWh (Ates et al., 2020). Furthermore, a study in Ref. Goswami and Kumar (2021) implemented a 10 MW FSPV project, they considered the technical and economic parameters to perform the feasibility analysis. The results showed that the FSPV system could generate approximately 10.2% more electric power than land-based PV plants system. As a result, the levelized-tariff cost of FSPV is then reduced to 39% than other types of PV plants.

Moreover, several authors have rationally applied various model techniques to assess the technical, economic and environmental feasibility of installed floating solar PV (FPV) and hybrid systems power generation performances in diverse countries, across the

globe. Some of these notable related studies reviewed, are summarized and presented in Table 1. The literatures show there have been many studies regarding individual reservoirs/dams, ponds and lakes, and supported by different models on floating PV systems performance with significant interest, as effective means to meet increasing energy needs for industrialization.

Despite the benefits, few studies such as Gonzalez Sanchez et al. (2021), Moner-girona et al. (2021) conducted review works assessing floating solar photovoltaics potential in existing hydropower reservoirs in Africa. Little research has been conducted in the Sub-Saharan Africa, particularly in countries like Ghana where utility-capacity floating PV systems will be installed in the long term. Henceforth, the PV potential of available water-bodies in Ghana should be assessed. The already existing studies for Ghana focused mainly on land-based PV performance technologies (Aboagye et al., 2021a). Although, in Ghana, there is an installed 5MW floating solar plant, which forms part of a 250 MWp solar energy generation project at Bui hydropower site, making it the first to be commissioned in the sub-region, and Ghana's first hybrid plant utilizing both solar and hydro resources to generate and supply power to the national grid (Bui Power, 2019). This hydro-solar PV hybrid (HSH) system within the Bui enclave is aimed at augmenting the country's peak hours and preserving the Bui reservoir. Despite Bui floating PV installed project, there are no feasibility studies on FPV generation performance in the open literature for Ghana that focus on employing hybrid solar FPV-hydro systems for electricity generation.

Ghana, a country located in Sub-Saharan Africa, experiences significant levels of solar radiation throughout the year, hence, has excellent potential for FPV installation on other two of its hydropower reservoirs (Akosombo-Black Volta dam reservoir, and Kpong river) and also to expand the existing Bui floating solar generation on the Black Volta. In spite of all this, the country heavily depends on fossil-fuels sources to generate electricity to meet the countries-based load. Resulting in high energy costs, toxic carbon emissions. Additionally, vast farm lands are also wasted for PV plant installations. Currently, the installed RE capacity in the country stands at 42.6 MWp as at June 2020 (Energy Commission of Ghana, 2020a). In view of this, the government of Ghana launched a campaign with policies and programmes to explore renewable energy projects in with the target of increasing Renewable Energy (including FPV systems) mix by 10% by 2030 to help achieve the Sustainable Development Goals of the United Nations. Research has shown that when 3%–4% hydroelectric dam reservoirs area are used for solar power generation, it significantly improve the generation mix and help reduce energy costs and improve PV sustainability (Energy Commission of Ghana, 2020b). The Akosombo Hydropower station is located on Volta Lake with a surface area of 8500 km². Yet, FPV-hydro technology on the Akosombo Volta lake has not been exploited. Adopting this hybrid system technology will significantly enhance the cost efficiency of solar energy in the country with upfront benefits to the nations clean energy campaign, addressing climate change and its hostile impact on socio-economic growth. In view of this, assessing this feasibility can provide insights into how effectively solar irradiation and dam reservoir areas translates into useful electricity, to provide a vital benchmark for FPV system optimization.

Therefore, this study introduces RETScreen Expert and Google Earth program to evaluate the feasible PV electricity performance from specific water body coverage respectively. The purpose of this study is to evaluate the feasible applicability and potential of floating PV on the large Akosombo hydroelectric dam reservoir in Ghana.

This study's main objective is to conduct technical, environmental and financial opportunities of integrating FPV into the

Table 1

Summary: Related studies on model techniques for FPV plants performance.

Author(s)	Detail of the study	Method/ Technique type	Location
Liu et al. (2017)	The study examined the effect of cooling and temperature variation on the FPV performance. According to their findings, about 160 GW of power can be generated from a reservoir with 2500 km ² surface area.	finite element model	China
Liu et al. (2019)	These studies proposed and analysed, a combined operation of a 2 GW FSPV plant and 1 GW pumped storage power (PSP) system performance to achieve maximum power efficiency and minimum power imbalances. The experiment operation outcomes illustrated an improved power output as 9112.74 MW and the reduced energy imbalance as 23.06 MW.	Genetic Algorithm	China
Oliveira-pinto and Stokkermans (2020)	This study analysed FSPV performance implementation using simulation models to analyse the technical and economic feasibility. Results from the study showed 1.81%–2.59% increase in power output and significantly reduced costs.	PVsyst [®] model	Brazilian
Golroodbari et al. (2021)	The study performed a Techno-economic analysis of a hybrid FSPV plant and a wind farm incorporating cable pooling to enhance solar capability. The study outcome shows that the combined operation lessens cost with improved efficiency. They also worked on solar and wind resources' effects on power generation.	Mathematical model	Netherlands
Yashas and Aman (2022)	This study analysed floating solar photovoltaic (FSPV) technology to harness sustainable energy, considering 32 lakes spanning across 3294 ac of lake area, assess solar electricity generation output and estimate water evaporation losses. The study shows that the FSPV systems with a coverage ratio of 0.5–0.6 could meet an average of 26% of the city's annual power demand.	RETScreen, PVWatts, PVGIS	India
Sukarso and Kim (2020)	In this study, the authors predicted FPV efficiency and measured energy yield from the system while also developing an economics analysis on an FPV project by comparison with ground-based solar PV (GPV). The study results show the lake has a cooler temperature than the ground, with an annual difference of around 8 °C. FPV efficiency was also shown to be around 0.61% higher than GPV in terms of the prediction. FPV economic parameter comparison also resulted in 3.37 cents/kWh lower levelized cost of electricity (LCOE), and 6.08% higher internal rate of return (IRR) compared to GPV in the base scenario.	remote sensing method SAM software	Indonesia
Semeskandeh et al. (2022)	This study evaluated a techno-economic-environmental feasibility performance to construct a 5-kW FPV and ground PV (GPV) power plant. Additionally, the FPV system is compared with the ground PV. Effects of wind and water temperature were considered in the model. Also, a sensitivity analysis was performed due to the uncertainty in climatic conditions. The simulation results showed an FPV system production capacity and panels' efficiency of 19.47% and 27.98% higher than the those of the GPV system. In addition, the FPV system was found to have a 16.96% increase in the annual performance ratio. Overall, the FPV system reduces the equity payback to 6.3 years (a 22.2% reduction compared to the GPV power plant).	MATLAB Simulink and RETScreen [®] software	Iran
Febrian et al. (2023)	This study performed Floating PV power plant performance with power capacity of 145 MWp Cirata Reservoir using different types of solar panel. The study results showed an annual energy capacity for monocrystalline, polycrystalline, and thinfilm solar cell module types as 190.54 GWh, 190.54 GWh, and 186.87 GWh respectively. With capacity factors of monocrystalline, polycrystalline, and thin-film solar cell module types respectively of 15.00%; 15.00%; and 14.70%.	RETScreen Expert model	Indonesia
Mittal et al. (2017)	This study evaluated the feasibility performance of installing FPV system at Ana Sagar Lake (Ajmer), Kaylana Lake (Jodhpur), Kishore Sagar Lake (Kota), and Man Sagar Lake (Jaipur). From the work AC energy produced by FPV system, by covering 5%, 10%, 15%, and 20% of lake's area are calculated for each of the lakes. Additionally, savings in water by reduction in evaporation due to FPV estimated. The outcome of the study shows that the potential of FPV system is highest for Man Sagar Lake (Jaipur) followed by Kaylana Lake (Jodhpur), Ana Sagar Lake (Ajmer), and least for Kishore Sagar Lake (Kota).	RETScreen Expert model	India.

Akosombo hydroelectric reservoir on the Volta Lake in Ghana, for energy savings, systems reliability, financial savings, and GHG emissions mitigations in Ghana. Additionally, this study can contribute to feed-into the target of developing sustainable energy-mix policies by 10% by 2030 to promote RE sources in Ghana.

The study's remaining sections are as follows: Section 2 presents the materials and methods and employed to achieve the research goal. Likewise, Section 3 highlights on the results and discussions, then Section 4 provided conclusions comments on the study's key findings.



Fig. 1. Akosombo hydropower dam and reservoir site in Ghana (Google Earth, 2021).

Table 2
Dam site geological data.

Parameter	Unit	Climate data value
Latitude	°N	6.1
Longitude	°E	0.1
Elevation	m	19
Heating Design Temp.	°C	20.7
Cooling Design Temp.	°C	31.0
Climate zone	–	0A – Extremely – hot – Humid

2. Materials and methods

This section highlights the approaches used to conduct the technical, economic, and environmental (emissions) analyses of the hybridized FPV – hydro systems. Also, the methodologies used for estimating the feasible reservoir coverage area are explained.

2.1. Study site description

Akosombo Hydropower Plant (AHP) dam site is the study area, and with a capacity of about $153 \times 10^9 \text{ m}^3$, and it provides different services for the inhabitants around; from fishing to navigational purposes. The area is also having 4.66 kWh/m^2 of daily solar energy (Fiabge and Obeng, 2006; Gyamfi et al., 2015). The facility provides about 38.3% of the bulk of Ghana's electrical power. The feasibility of the floating solar PV-hydro system was conducted for Akosombo dam site using satellite data recorded from RETScreen database. Table 2 indicates the dam Site geological data (Latitude 6.1°N , Longitude 0.1°E). additionally, Google Earth online engine view of the study site presented in Fig. 1 (Google Earth, 2021).

2.2. Akosombo hydropower plant capacity and climatic data of study site

The station currently has electric-power-generating capacity of 1020 MW. The generation capacity at 2020 for hydro, solar and thermal generation was 5137 MW which is about 12.2% more than in 2018 (Ghana Grid Company Limited, 2020). Fig. 2 shows the hydroelectric power generation from 2017 to 2020.

Hydroelectric power generation heavily depends on water levels and flow in the dams. From Fig. 2, it can be deduced that the yearly electricity generation was not consistent and creates huge energy deficit gap, hence the need for a hybrid solar PV-hydro system to serve as a sustainable energy fill to this energy gap. This yearly electricity generation deficit challenge can be largely attributed to weather instability, such as high daily temperatures, contributing to evaporation reservoirs. The propose hybrid system uses abundant solar radiation and the waterbodies available to produce energy (IEA, 2019). As at the time of the study there was no environmental information about the location. To resolve this situation, RETScreen Expert software (Anon, 2023) was accessed to retrieve recent solar resource data for Akosombo Dam site. According to Sinha and Chandel (2014), the RETScreen Expert provides full meteorological data based on specific climatic location. Table 3, presents NASAs updated Near Real-time Global Radiation and Meteorological data assessed from RETScreen Expert for Akosombo.

The magnitudes of the values indicate excellent meteorological climate conditions for PV energy generation (Aboagye et al., 2021b) with an average yearly radiation of $4.66 \text{ kWh/m}^2/\text{day}$. In addition, Table 3 indicates the monthly average variance in relative humidity (Hr). The Hr values reported declines throughout the hot weather season between November–April, reaching 63.9 percent in January. With the onset of the rainy seasons (May–October), Hr rises once more, peaking at 85.6 percent in June and July. At the Akosombo Hydro site, the yearly average of Hr is 77.9%. In this study, the climate data shows that when Hr is 65.5 percent, the ambient air temperature reaches 28.6°C with 3.0 m/s air speed, depicting better climatic conditions for PV installations.

2.3. The RETScreen modelling

The Renewable Energy Technology Screen (RETScreen) software (Chowdhury et al., 2020b) was selected to help carry out table top simulations for decision for system sizing the energy generation system. The tool was chosen because of its extensive applications for clean energy decision making studies This simulation application is a one-of-a-kind decision-support tool for examining energy generation, pricing, reduction of greenhouse

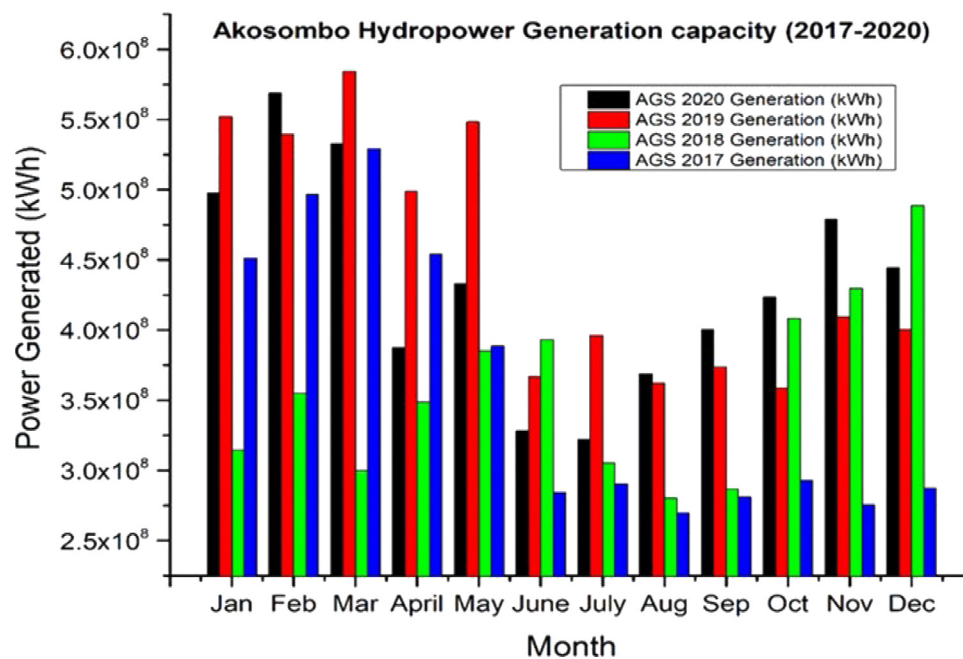


Fig. 2. Akosombo hydropower generation capacity between (2017–2020).

Table 3

Solar and climate data of Akosombo Volta Lake site.

Month	Temperature (°C)	Relative humidity (%)	Daily solar radiation- GHI (kWh (m ² d) ⁻¹)	Wind speed (m s ⁻¹)
January	27.6	63.9%	4.08	2.5
February	28.6	65.5%	4.78	3.0
March	28.1	72.9%	5.06	3.1
April	27.6	78.3%	5.30	3.1
May	26.7	82.7%	5.15	2.9
June	25.5	85.6%	4.67	3.0
July	24.5	85.6%	4.03	3.5
August	24.5	83.2%	4.14	3.6
September	25.1	83.6%	4.73	3.0
October	25.7	84.2%	5.11	2.6
November	26.7	78.8%	4.84	2.3
December	27.3	69.9%	4.06	2.2
Average	26.5	77.9%	4.66	2.9

gas, financial feasibility, risk of different resource combinations for any place on the globe (Chowdhury et al., 2020b). RETScreen is a Windows-based energy management software program for evaluating energy system performance.

In this study, RETScreen expert was applied to estimate the FPV plant feasibility performance for technical, economic and environmental on the Akosombo dam reservoir. Simulations were performed to determine the quantity of PV panels that will fit the feasible coverage area for an appropriate FPV design and the electricity generation capacity. Climatic data were accessed for the feasibility performance through RETScreen Expert NASA database with integrated electricity generated power plants. Additionally, RETScreen performed a Monte Carlo simulation for comparative risk analysis to benchmark FPV source unit energy production cost to existing conventional energy sources, to ascertain FPV systems cost-effective production performance against the plant capacity. A unit energy production cost of 0.10 \$/kWh from RETScreen simulation was used for the benchmark analysis, as shown in Fig. 3.

2.4. PV module reservoir coverage assessment

Safe zone within the dam boundaries with adequate depth that is suitable for the stationing of the panels before attaching

them to their accessories were assessed on the Volta Lake. Selection of portions on the water bodies that can accommodate the PV panel is of critical importance especially the floating structure like ponton, mooring lines and floaters. Assessment was done using images captured and calculated from Google Earth online engine (Ismail et al., 2022).

In this study, Google earth program was used to assess and illustrate the energy production capacities per considered surface areas on the Akosombo reservoir. The potential of floating solar PV energy production at Akosombo dam site were investigated considering less than 0.1% (<0.1) reservoir coverage area. This study, then uses RETScreen Expert software and applied the sub-systems (solar panel, inverter, transformer) input parameters into the model to estimate the reservoir surface area applicable for a certain electricity generation. Then, Google Earth software was used to demonstrate the effective reservoir coverage area as shown in Fig. 4. The Google Earth online program considered the Akosombo reservoir depth (maximum lake level: 84.73 m and minimum lake level: 73.15 m) as a reference input levels for the underwater ground range, then to perform the water depth mapping simulation. The simulation was conducted to select feasible reservoir areas with depths corresponding to the maximum and minimum depth range to withstand the simple water variations as well as flood water levels with no obstacles beneath. This

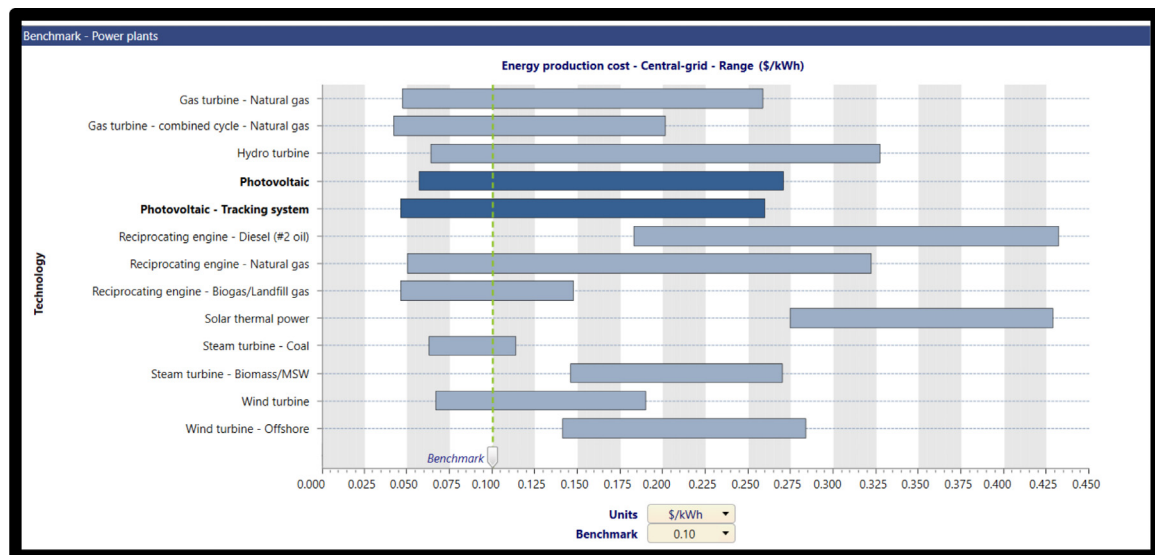


Fig. 3. Benchmarks of unit energy production cost (\$/kWh) of proposed FPV system to other energy sources RETScreen expert.



Fig. 4. Feasible reservoir areas (Yellow Boundary) of 2.46 km² assessed for the FPV plant capacity in the study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

was to ensure the system is able to withstand minimum sag, considered for the floating PV systems anchoring to the waterbed, and avoiding the floating bed sinking or sagging towards the shore to cause damage to the PV system and power cables. This online tool has specific characteristics for gauging the underwater ground elevation of a reservoir and measuring border layouts on water bodies. The software helped find the feasible. The feasible simulated reservoir size with adequate depth to install the PV panels was 2.46 km², as highlighted in Fig. 4.

2.5. Modelling FPV-hydro plant system

Design parameters presented in Fig. 5, were analysed to measure the possibility of integrating potential renewable resource like floating PV system into the hydro power station. The technical performance, economic and environmental impact models

were then applied to obtain relevant output performance variables including the annual energy yield, the levelized cost of electricity (LCOE), simple payback time (SPBT), net present value (NPV), GHG CO₂ and water savings, and internal rate of return (IRR). The floating PV power output was computed using design factors such as solar irradiation, PV panel characteristics (tilt angle, module efficiency, module cell temperature, and module power), inverter efficiency, and panel installation area in RETScreen Expert.

Furthermore, extra hydroelectric power from saved evaporated water was estimated using main hydropower design parameters from Akosombo hydro plant catalogue and adopted available data including dam water level, water flow rate, turbine efficiency, available hydraulic height, reservoir volume capacity, average discharge ratio of the reservoir (Kougias et al., 2016).

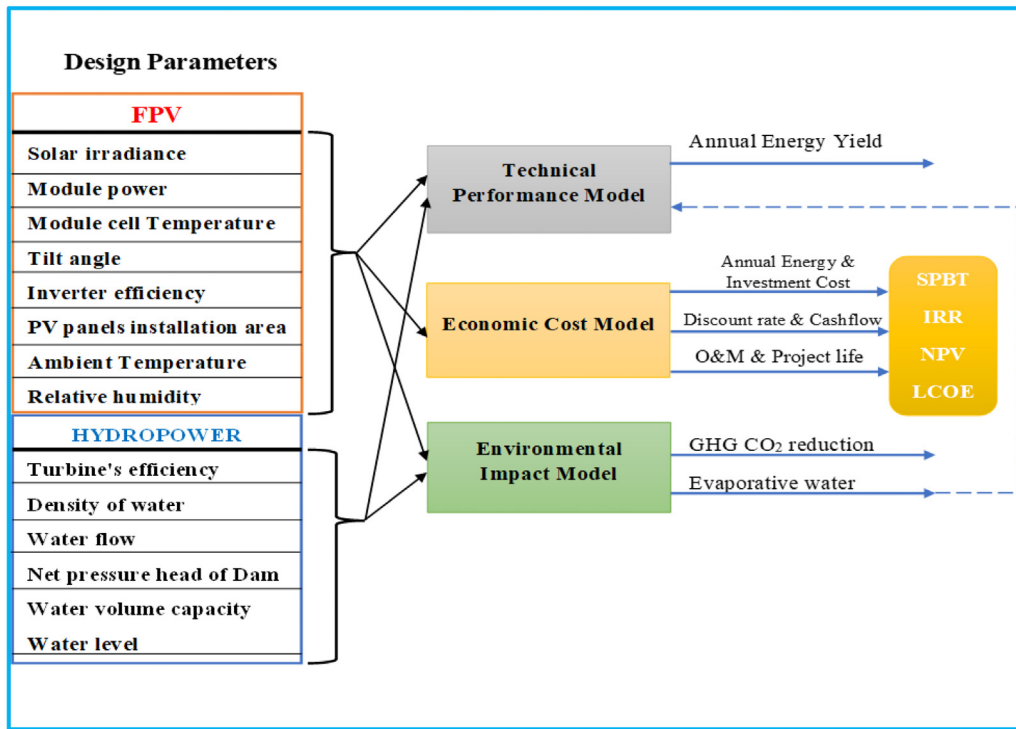


Fig. 5. Design parameters of FPV-Hydropower hybridized plant.

Fig. 5 presented the design parameters implemented in this study. The factors that affect FPV output performance including ambient temperature, irradiance levels, cell temperature, panel installation coverage area etc. were assessed. The performance indicators include solar PV energy output and evaporation.

2.5.1. Modelling solar PV output power performance

RETScreen computes the PV array power output using Eq. (1), based on total solar irradiance at the selected dam site as follows (Murat et al., 2020);

$$P_m = G_t A_m \eta_{\text{system}} \quad (1)$$

P_m , G_t , A_m , η_{system} denote module (panel) output power (kW), total irradiance (global radiation on the horizontal surface), covered area of floating PV module (panels) and PV system efficiency. The FPV plant system efficiency is estimated using Eq. (2), (Murat et al., 2020);

$$\eta_{\text{system}} = \eta_m * \eta_{\text{inverter}} * \eta_T * \eta_{\text{cable}} \quad (2)$$

η_m , η_{inverter} , η_T and η_{cable} denote module efficiency, inverter efficiency, transformer efficiency and cable and junction box efficiencies respectively. A collective value of 0.95 was used to represent η_{inverter} , η_T and η_{cable} (Murat et al., 2020), while η_m was 0.15 for a silicon PV panels. Moreover, for temperature-dependent FPV module efficiency (%), Eq. (3) is used to estimate efficiency of a PV module (Murat et al., 2020);

$$\eta_m = \eta_{m, \text{stc}} [1 - \alpha_t (T_c - T_a)] \quad (3)$$

α_t , T_c , T_a , $\eta_{m, \text{stc}}$ denote PV module temperature coefficient, module temperature of the module, temperature of the PV module at $T = 25^\circ\text{C}$ and efficiency of the PV conversion due to the influence of the deviation of the PV temperature in relation to the STC values. Here, module cell temperature (T_c), is estimated from Eq. (4), (Murat et al., 2020);

$$T_c = T_{\text{amb}} + \frac{(NOCT - 20^\circ) * G_t}{0.8} \quad (4)$$

T_{amb} , $NOCT$ and G_t denote ambient air temperature, average design operating cell temperature, at $45 \pm 2^\circ\text{C}$ ($I = 800 \text{ W/m}^2$, $T = 20^\circ\text{C}$ @ $m_{\text{air}} = 1.5$) and total irradiance. Table 4 shows the characteristics of the name plate of selected panel.

2.5.2. Evaporation reduction calculation

Reduction in water through evaporation from a reservoir is a major benefit of FPV to the reservoir (Eshun and Amoako-Tuffour, 2016). Several environmental and climatic parameters are necessary when estimating the evaporation rate (E_r) from Eq. (5) (IRENA, 2019).

$$E_r = 0.047 R_f \sqrt{9.5 + T_{dm}} - 2.4 \left(\frac{R_f}{R_{et}} \right)^2 + 0.09 (20 + T_{dm}) (1 - 0.01 H_r) \quad (5)$$

E_r , T_{dm} , R_{et} , R_f and H_r denote average daily water surface evaporation (mm/day), mean daily temperature ($^\circ\text{C}$), extra-terrestrial radiation ($\text{MJ/m}^2/\text{s}$), radiation flux ($\text{MJ/m}^2/\text{s}$) and relative humidity.

The amount of water stored as a result of effect of reduced evaporation (ΔV) (m^3) is estimated by Eq. (6), (Farfan and Breyer, 2018).

$$\Delta V_w = A_{\text{FPV}} * \left(\frac{E_r}{1000} \right) \quad (6)$$

A_{FPV} denotes surface area of panel covered on water (m^2).

W_p (kWh) denotes the maximum electricity energy of the power plant (kWh) that could be generated by the hydroelectric using the volume of water saved (m^3) from evaporation, derived from Eqs. (7) to (10), (Farfan and Breyer, 2018).

$$W_p = \eta * \rho * g * H * Q * T \quad (7)$$

$$T = \frac{\varphi}{\theta} \quad (8)$$

W_p , η , ρ , g , H , Q , T , φ , ΔV_w and θ denote maximum electricity energy of the power plant (kWh), hydraulic turbine efficiency and

Table 4
Characteristics of the inverter and the PV system.

Solar module	Inverter type
Manufacturer: Q-Cells	Rated Power Capacity: 90, 000 kW
Model: mono-Si - Q. PEAK BLK-G4.1- 285 W	Vmax_mppt: 480 V
Rated power: 285Wp (mono-crystalline silicon)	Inverter DC to AC ratio: 0.99
Nominal Efficiency (η): 17.07%	Efficiency: 95%
Operating Cell Temperature (NOCT): 45 °C $\pm$ 2 °C	Miscellaneous Losses: 1%
Temperature Coefficient: 0.4%	
Miscellaneous Losses: 16%	
Life Expectancy: 25 years	
Panel Frame: 1.67 m ²	

electricity generator in %, (85% was chosen [Sinha and Chandel, 2014](#); [Kougias et al., 2016](#)), density of water (1000 kg/m³), acceleration of gravity constant, effective net pressure head of the dam (m), Water flow (m³/s), time in hours, reservoir capacity in m³, volume of water saved by evaporation (m³) and mean annual reservoir discharge (m³/s) respectively.

2.5.3. Economic performance modelling (financial and cost analysis)

Energy payback time (EPBT), levelized cost of energy (LCOE), net present value (NPV), and internal rate of return (IRR) models were utilized in the economic performance model. The simple payback time (SPBT) present the amount years required for cash flow (except. loan payments) from using FPV energy to match total investment (equal to aggregate debt and equity), presented in Eq. (9) as [Durkovic and Djuricic \(2017\)](#) was used to derive the basic payback period:

$$\text{Payback (years)} = \frac{C_i (\$)}{(100\% - \text{Tax}(\%)) * P_w \left(\frac{\text{kWh}}{\text{year}} \right) * y_n \left(\frac{\$}{\text{MWh}} \right) - O\&M \left(\frac{\$}{\text{year}} \right)} \quad (9)$$

C_i , P_w , y_n and $O\&M$ denote initial cost (\$), Annual FPV power production, energy charge ($\frac{\$}{\text{MWh}}$) and Operational and Maintenance cost ($\frac{\$}{\text{year}}$).

When applied to project cash flows, the internal rate of return (IRR) is the discount rate which results in a net present value (NPV) of zero (NPV = 0). The expected project performance can then be calculated with that discount rate. The project is approved if the IRR is more than a predetermined percentage. Project is rejected when IRR is below target ([Chowdhury et al., 2020b](#); [Ismail et al., 2022](#)).

To calculate the IRR value, Eq. (10) was utilized.

$$\text{IRR} = R_a + \frac{\text{NPV}_a}{\text{NPV}_a - \text{NPV}_b} (R_b - R_a) \quad (10)$$

IRR, R_a , R_b , NPV_a and NPV_b denote the internal rate of return, lower discount rate chosen, and higher discount rate chosen, NPV @ R_a and NPV @ R_b .

The net present value for the project is computed discounting all cash streams using Eq. (11), ([Arjunan, 2019](#)) as:

$$\text{NPV} = \sum_{t=0}^n \left(\frac{CF_t}{(1+r)^t} \right) \quad (11)$$

CF_t , r , t and n denote cash flows in the period (n), rate of return, time period and total number of periods, respectively. After removing the current investment, the net present value (NPV) is the current amount of future cash flows returns on investment that have been discounted to their current value. The target rates put the net present value as close to zero as possible may be attained by applying the NPV formula and filling in rates ([Drury et al., 2011](#)).

The LCOE of the FPV was calculated based on the simplified LCOE Eqs. (12) and (13), ([Huld and Commission, 2014](#); [Chowdhury et al., 2020a](#)):

$$\text{LCOE} = \frac{\sum_{t=0}^n \frac{I_t + M_t + E_t}{(1+(1+R\%)^t)}}{\sum_{t=0}^n \frac{E_t}{(1+R\%)^t}} \quad (12)$$

$$\text{LCOE} = \frac{\text{sum of discounted costs over lifetime}}{\text{sum of discounted energy provided over lifetime}} \quad (13)$$

I_t , M_t , F_t , E_t and $R\%$ denote capital spending in year t (\$), O&M spending in year t (\$), fuel spending in year t (\$), energy production in year t (MWh) and discounted rate (%), when the t = projected lifetime of the project system (in yrs.).

2.5.4. Environmental impact modelling (GHG emission mitigation potential)

The effect of the FPV system on greenhouse gas reductions. (GHG) emission was calculated using Eq. (14) ([Durkovic and Djuricic, 2017](#); [Diawuo et al., 2020](#); [Anane-Fenin, 1986](#)):

$$G_r \left(\frac{t\text{CO}_2}{\text{MWh/year}} \right) = P_w * G_s * (1 + \beta) \quad (14)$$

G_r , G_s , P_w and β denote annual reduction of GHG, $\left(\frac{t\text{CO}_2}{\text{MWh/year}} \right)$, standard value of GHG emissions for Ghana, $\left(\frac{t\text{CO}_2}{\text{MWh}} \right)$, annual energy production $\left(\frac{\text{MWh}}{\text{year}} \right)$ (total potential energy of FPV + AHPD) and a dimensionless parameter of transmission and distribution systems average loss.

3. Results and discussion

The study's findings are presented and discussed in this section. It highlights the FPV system's technical performance, financial and economic analysis and GHG emission savings. Additionally, evaporation savings for electricity generation feasibility was also assessed.

3.1. Electricity generation of FPV system

RETScreen Expert modelling program was used to size and analyse a feasible FPV plant system through multiple input parameters, with the assumption of occupying less than 0.1% (<0.1) of reservoir area, a location with an average annual solar radiation of 4.66 kWh/m²/day. The analysis types and methods chosen in RETScreen are mini grid and method two respectively From [Fig. 6](#), RETScreen simulation input parameters considered includes; 285 Wp (Q-Cells manufactured) mono-crystalline silicon cell was selected for the investigation. Detail information about the solar cell and inverter specifications is shown in [Fig. 6](#). Results from FPV system modelling established a plant capacity factor of 14.1% for the solar collector area of 2,460,457 m² (2.46 km²). An annual amount 520,223 MWh is produced into the grid from FPV at a rate of 0.10 \$/kWh. This amount of energy requires a total of

RETScreen - Energy Model Subscriber: UENR - Educational Use Only

Power plant - 420,000 kW - Tracking system - Photovoltaic

Fuels & schedules

- Electricity and fuels

Technology

- Power
 - Photovoltaic - 420,000 kW - Tracking system
 - Photovoltaic

Summary

- Include system?
- Comparison

Photovoltaic

Description: Photovoltaic - 420,000 kW - Tracking system

Note:

Level: Level 1, Level 2, eLearning

Photovoltaic - Level 2

Resource assessment

Solar tracking mode: Fixed

Slope: 6

Azimuth: 0

Show data

Photovoltaic

Type: mono-Si

Power capacity: 420,000 kW

Manufacturer: Q-Cells

Model: mono-Si - Q-PEAK BLK-G4.1 - 285W

Number of units: 500,000

Efficiency: 17.07%

Nominal operating cell temperature: 45 °C

Temperature coefficient: 0.4% / °C

Solar collector area: 2,460,457 m²

Bifacial cell adjustment factor: 0%

Miscellaneous losses: 16%

Inverter

Efficiency: 95%

Capacity: 90,000 kW

Miscellaneous losses: 1%

Summary

Capacity factor: 14.1%

Initial costs: \$/kW 1,350

O&M costs (savings): \$/kW-year 12

Electricity export rate: Electricity export rate - annual

Electricity exported to grid: MWh 520,223

Electricity export revenue: \$ 52,022,343

Fig. 6. PV system and inverter input characteristics and parameters in RETScreen expert.

60 arrays, each with a 7 MWp power capacity from 500,000 PV panels (1.67 m² frame area) to be placed within the calculated area of 2.46 km². A total of 420 MWp power was attained for the system as each array has 7 MWp. Relatively, the estimated 420 MWp FPV energy is about 10% more than Ghana's installed generation capacity from REs in Ghana, at the end of June 2020, which stood at 42.6 MW. The proposed installed capacity in this study is comparatively 41% of the 1020 MWp Akosombo hydroelectric plant capacity. Moreover, results in Table 5 shows the total monthly radiation on a horizontal surface, and the corresponding electricity exported to the grid-network with the export rate of the generated electricity.

From Table 5, the average irradiance of 4.66 kWh/m² throughout the year on the Global horizontal irradiance (GHI), with the lowest and highest concentrations at 4.035.30 kWh/m² and 5.30 kWh/m² respectively occurred on July and April followed by October. In April, when solar irradiance is high, the FPV system delivered 47,459.774 MWh of electrical energy to the grid, and 37,800.361 MWh in July. For 0.42 GWp FPV solar power plant, 520,233.433 MWh is exported into the grid each year. These findings are comparable with those obtained by Mittal et al. (2017) in

India, Oliveira-pinto and Stokkermans (2020) in Brazil, Liu et al. (2017) in China, and Yashas and Aman (2022), Semeskandeh et al. (2022), Febrian et al. (2023) in India, Indonesia Iran respectively, where using FPV in electricity generations with RETScreen model.

The monthly average daily total of energy delivered to grid by the 420 MWp floating PV system, as well as the monthly maximum irradiance statistics, are detailed in Fig. 7. Certainly, Fig. 7, confirms that, output power of solar module increase as irradiance increases, which observed the highest electricity generation accruing in October, and lowest in July during the raining season. Henceforward, solar PV systems will export 12, 875.77 GWh FPV power into the grid during a 25-year period, considering 1% system degradation rate (Chowdhury et al., 2020a).

3.2. Cost analysis of FPV system

Appropriate estimations of cost and financial characteristics such as the initial cost, yearly capital cost, O&M cost and revenues, as well as debt payments are analysed in order to assess the proposed system's expenses. They are critical in determining

Table 5
Average monthly FPV electricity exported to the grid.

Month	Daily solar radiation – GHI (kWh/m ² /d)	Electricity export rate (\$/kWh)	Electricity exported to grid (MWh)	Electrical energy exported to grid (GWh)
January	4.08	0.10	39,877.035	39.88
February	4.78	0.10	41,294.962	41.29
March	5.06	0.10	47,640.334	47.64
April	5.30	0.10	47,459.774	47.46
May	5.15	0.10	47,190.456	47.19
June	4.67	0.10	41,633.695	41.63
July	4.03	0.10	37,800.361	37.80
August	4.14	0.10	39,171.859	39.17
September	4.73	0.10	43,517.245	43.52
October	5.11	0.10	49,098.170	49.10
November	4.84	0.10	45,639.615	45.64
December	4.06	0.10	39,899.927	39.99
Annual	4.66	0.10	520,233.433	520.233

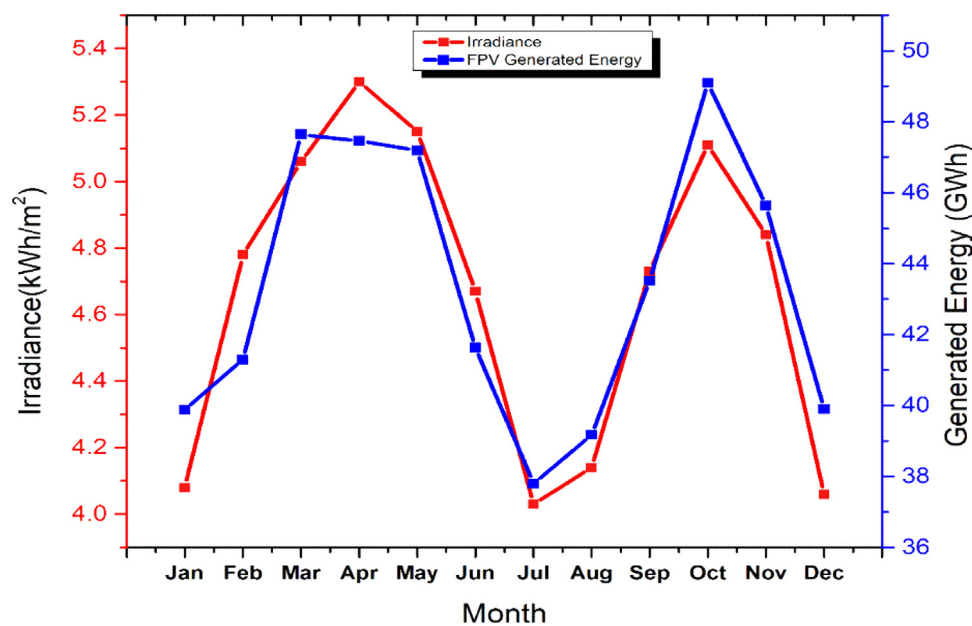


Fig. 7. Solar irradiation and generated energy and delivered by the FPV system to grid monthly.

the proposed project's economic viability. Conducting cost calculations using cost parameters in the RETScreen Expert model in order to evaluate project viability is vital part of project implementation to boost investors' confidence. The cost estimation presented in Table 6, indicates an initial cost of \$ 567,030,000.00 for the project with 99.99% of the cost coming from power system sources, acting as the major cost impact of the project. The power system sources include factors such as; PV panels, development, engineering, feasibility study, training and commissioning, inverters, cables, and accessories. The PV and inverter lifetimes were estimated to be 25 years in the proposed system, with a maximum debt repayment rate of 15 years. The overall yearly cost of the project is anticipated to be USD \$48,647,792.00. After all the expenses, the annual savings and revenue from the project is USD \$52,238,576.00, this includes benefits of GHG reduction revenue amount (USD \$216,233) (Table 6), demonstrating a viable, cost-effective and bankable and revenue generating project.

3.3. Financial analysis of FPV system

The RETScreen Expert Software allows an operator to enter a variety of financial scenarios with different data (discount rates), estimate crucial financial viability indicators (equity payback, simple payback, and Net Present Value (NPV)). The outcome for a typical financial scenario is represented in Table 7. The table,

Table 6
Initial and annual costs, annual savings of proposed FPV system.

Project initial costs summary		
	Initial costs (%)	Rate (USD \$)
Power system	99.99	567,000,000
Feasibility study	0.002	10,000
Development	0.004	20,000
Total initial costs	100%	567,030,000
Annual costs		
O&M costs (savings)		5,04,000
Engineering		20,000
Training and commissioning		8,000
Debt payments-15 years		43,579,792
Total annual costs (TAC)		48,579,792
Annual savings		
Electricity export revenue		52,022,343
GHG reduction revenue		216,233
Total annual savings and revenue		52,238,576
Net yearly cash flow – Year 1		3590784.18

shows the features that were employed in this study to conduct a 25-year studies with 2% inflation, which is comparable to average rates as of 2020 (Chowdhury et al., 2020b).

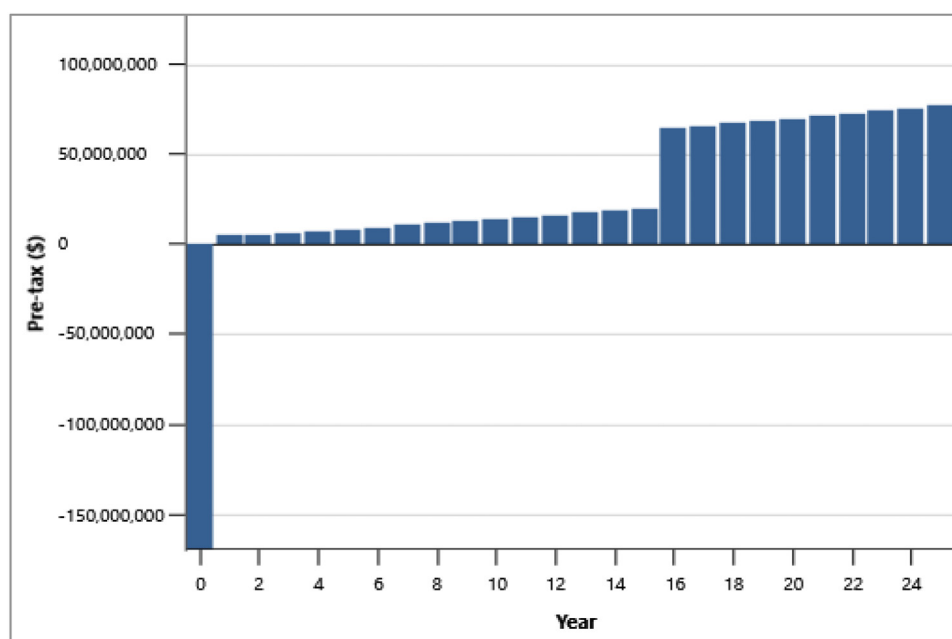


Fig. 8. FPV energy project annual pre-tax cash flows.

Table 7

Financial parameters in the proposed FPV project.

Financial parameter	Value (units)
Fuel cost escalation rate	1%
Inflation rate	2%
Discount rate	9%
Reinvestment rate	9%
Project life	25 Yrs
Finance	
Incentives and grants	\$30,000
Debt ratio	70%
Debt	\$396,921,000
Equity	\$170,109,000
Debt interest rate	7%
Debt term	15 yrs
Debt payments	\$43,579,792/yr
Simple payback	12 yrs
Equity payback	14.8 yrs
Net present value (NPV)	\$ 32,657,082
Annual life cycle savings	\$3,324,695
Energy production cost	\$/kWh 0.10

Similarly, at 9 percent discount rate, a yearly cost for O&M was estimated at 1 percent of capital cost (Chowdhury et al., 2020b). From the annual pre-tax and cash flow diagrams (Figs. 8 and 9), a positive cash flow was achieved after 14 years with a 12 years simple payback. It also demonstrated that, a positive Net Present Value (USD\$ 32,657,082) and annual life cycle savings of US\$3,324,695/yr., will be yielded, at an equity payback period 14.8 years. In terms of project economics, the FPV-Hydro hybrid plant system is cost-effective, since the system will generate bulk-profit return after 14 years of operation and reduce the system's total expenditures until the plant project life-cycles at 25 years. The annual pre-tax and cumulative cash flows of the FPV project are presented in Figs. 8 and 9 respectively.

3.4. GHG emission analysis of FPV project

For the GHG mitigation analysis, the GHG emission-based factor embedded in the RETScreen Expert database was used in this study's assessment. The emissions' analysis calculates the

proposed system's capacity to reduce greenhouse gas emissions (mitigation). The GHG emission reduction was estimated using RETScreen Expert, which analysed the proposed FPV case technology's annual emission reduction to Ghana's present fossil-energy technology's annual GHG emission reduction (base case scenario). 308,904.5 tonnes of CO₂ are saved from been released into the environment from generating 520,233 MWh yearly electricity at Akosombo dam site as presented in Table 8. The potential GHG emission reduction of the proposed FPV project decreases CO₂ emissions by 308,904.5 tCO₂/MWh when compared to the base scenario as shown in Fig. 10. This 308,904.5 tCO₂e is the same as 132,727,648.7 litres of gasoline not consumed. Comparatively, the emission mitigation tends to contribute immensely towards a greener and clean environment, and pegged at 2019 GHG reduction credit rate of \$0.70/tCO₂ for Ghana (Diawuo et al., 2020), an accumulated annual reduction revenue of USD \$216,233 was achieved. For instance, if the conventional energy sources are replaced with Floating PV systems in Ghanaian. These results are comparable with those attained by Febrian et al. (2023), and Semeskandeh et al. (2022) in India and Indonesia respectively, as deploying FPV would considerably reduce GHG emissions.

Financial analysis Deployment of FPV technology significantly will augment Ghana's ambitious SDGs climate commitments towards a clean energy agenda. From the analysis the hybrid system is more beneficial (CO₂ emission reductions, etc.) over the current thermal systems run by the country

3.5. Risk analysis of FPV system project

FPV project risk analysis was assessed to identify factors that could affect the project's economic life, utilizing all the parameters at a specific range to arrive at decisions. In this section, the RETScreen Expert software performed a Monte Carlo simulation considering 5000 number of combinations for the risk analysis, using energy production cost as the key financial indicator, at a $\pm 25\%$ range for the quantification of all factors. This produces a re-estimation of the energy production cost to produces the FPV impact risk analysis chart of the project as presented in Fig. 11. From Fig. 11, it can be observed from the chart that, the project

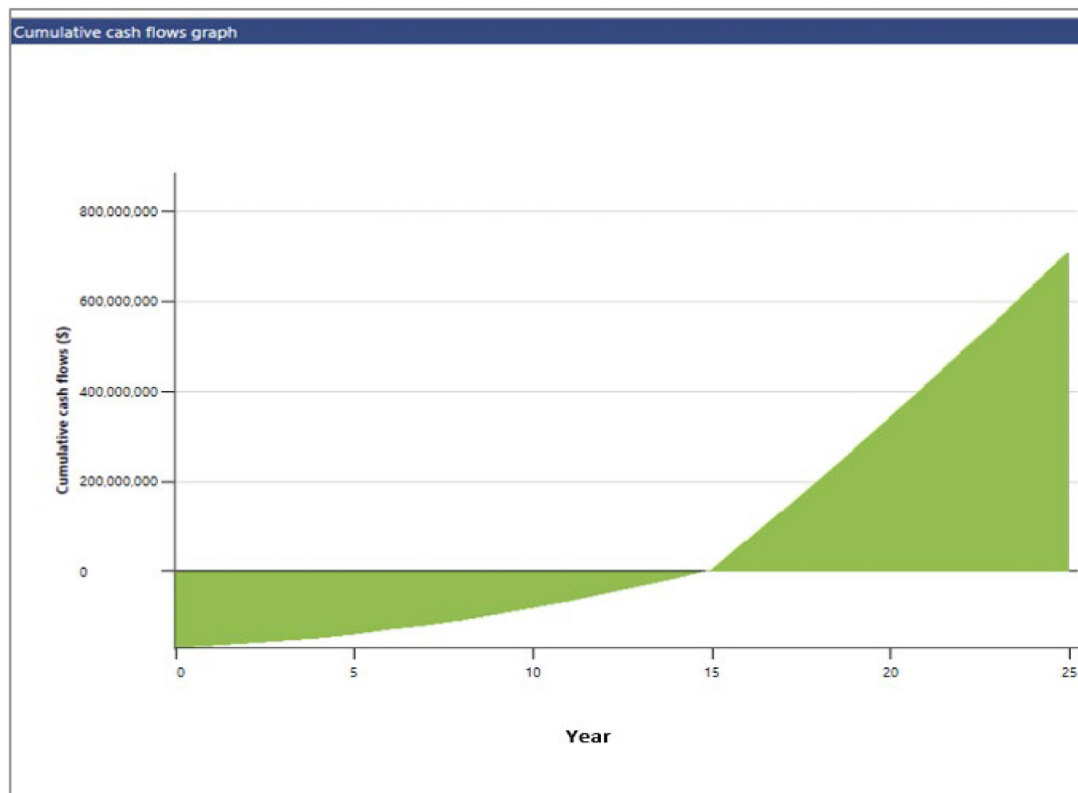


Fig. 9. Potential cumulative cash flow(s) from FPV systems project.

Table 8
GHG emissions reduction analysis.

Base Case system GHG Summary (Baseline)			
Fuel type	Fuel mix (%)	Fuel consumption (Ktoe)	GHG emission Factor tCO ₂ /MWh
Petroleum product	47.6	3794.0 (4409471.1 MWh)	0.638
Biomass (firewood/charcoal)	37.4	2981.3 (3464933.1 MWh)	
Natural gas (electricity)	15	1199.1 (1393620.7 MWh)	
Total	100%		332,155.3
Proposed Case system GHG Summary (Power proposed project)			
Fuel type	Fuel mix (%)	Fuel consumption (MWh)	GHG emission factor tCO ₂ /MWh
FPV solar	100%	520,233	0.007
Total	100%		23,250.9
Annual GHG emission reduction = 308,904.5 tCO₂			
NB: @GHG reduction credit rate of \$0.70/tCO ₂ , GHG reduction revenue = (USD \$216,233) for Ghana			

electricity exported to grid and initial cost significantly impacted the FPV project. Further deductions from the chart indicates a direct and an inversely proportional impact of the initial cost and electricity exported to grid to the cost of energy production respectively. Furthermore, in Fig. 12, the project risk level is set at 90% confidence level (10% risk level) generated the project distribution chart for risk analysis of the FPV system. Results in Fig. 12, showed the cost of energy production was 136 \$/MWh (0.136 \$/kWh) becoming relatively closer to the actual project cost of energy production of 100.0 \$/MWh (\$/kWh 0.10) attained before the risk impact assessment, confirming a low investment risk and bankable solar floating PV project.

3.6. Reduced evaporation analysis

Climate factors (wind speed, relative humidity and temperature) have a significant impact on the evaporation rate of a water body (reservoir). Typically, rate of evaporation is proportional

to the ambient air conditions (Goswami and Kumar, 2021). Evaporation rate at the Akosombo Hydro site was estimated. Furthermore, extraterrestrial radiation statistics and radiation flow data for Ghana was retrieved from Anane-Fenin (1986) and utilized in the evaporation rate computations. Fig. 13 depicts the monthly computed fluctuation in evaporation rate. The estimated evaporation rate increases when air temperature rises and at low Hr. At the Akosombo Hydro site, the annual average evaporation rate measures as 2.48 millimetres per day, and the maximum evaporation rate estimated at 3.08 mm/day in the month of February. Despite the moderate-to-high temperatures and wind speeds, the evaporation rate reduces significantly from May to October, due to the considerable rainfall at these times, there will be high Hr causing the air to have reduced moisture holding capacity then lowering the evaporation rate.

The total surface area of the FPV system was 2,460,457 m². A yearly evaporation reduction in the proposed 420 MWp FPV system saves 73,327 m³ of water per year (Fig. 13). Approximately

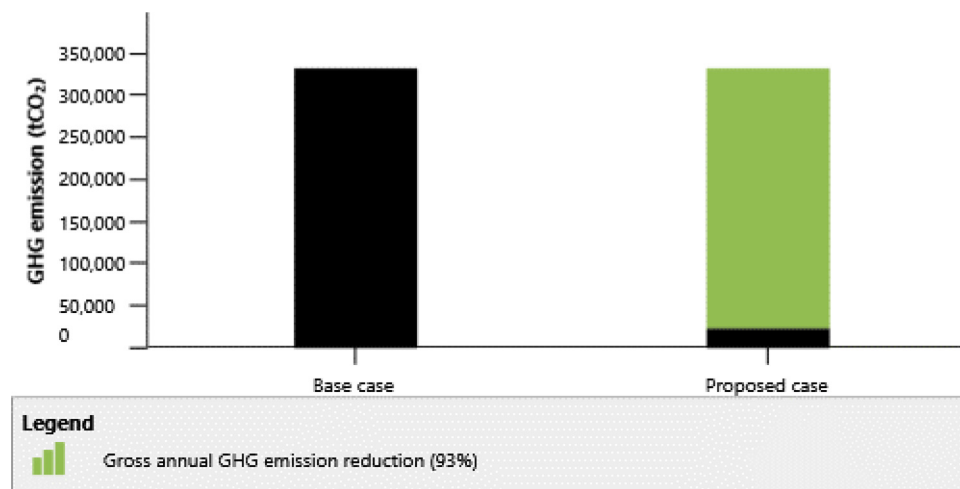


Fig. 10. Annual GHG emission mitigation potential of the proposed FPV systems.

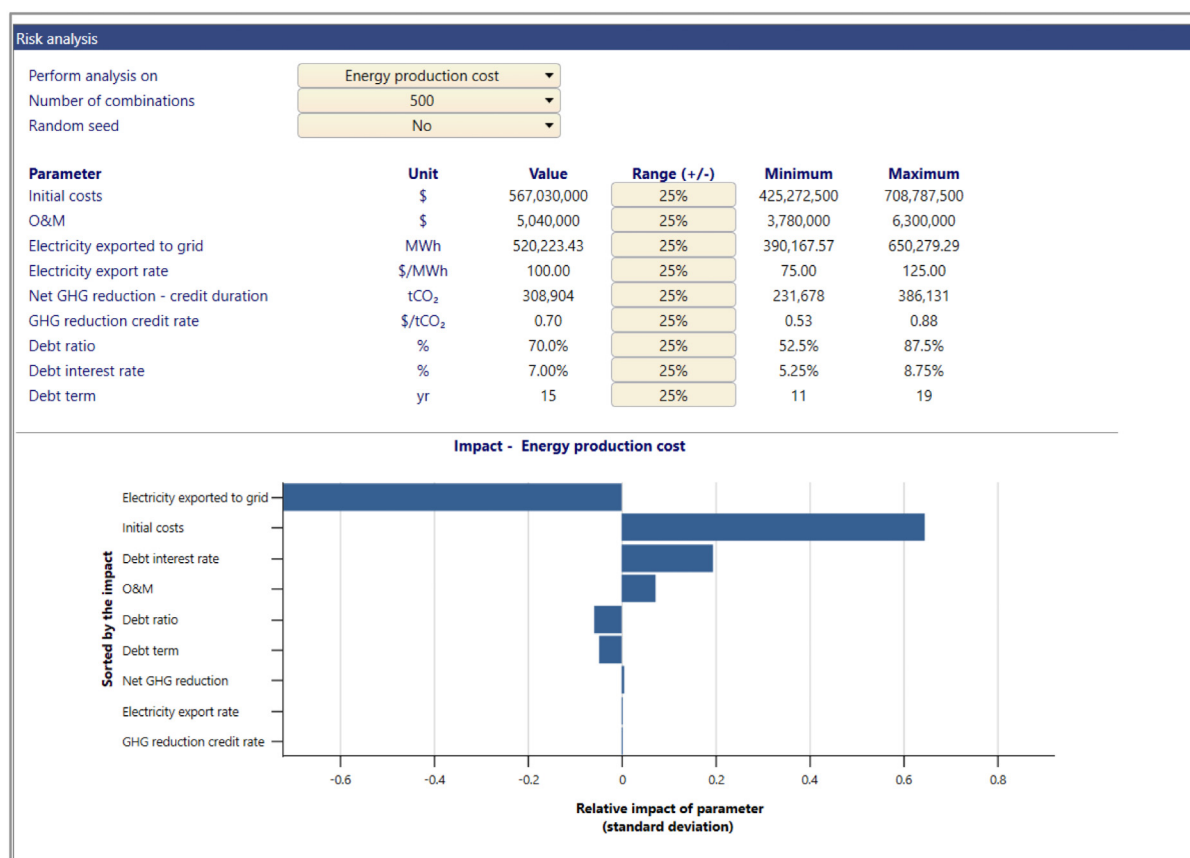


Fig. 11. Risk analysis impact chart on FPV project.

10 MWh of electrical energy would be generated from 73,327 m³ of total annual evaporation water saved (hydroelectric) and this is due to the shielding provided by the platform on a 2,460,457 m² dam surface. Thus, the FPV system will save 1,833,175 m³ of water during its lifetime (25 years).

4. Conclusion

In this work, a feasibility study was undertaken on an FPV-Hydro hybrid plant to establish the techno-economic and environmental viability of integrating an FPV plant into Akosombo

hydropower reservoir Dam to develop a 420 MWp FPV utility system. The findings are:

- RETScreen Expert simulations conducted at the Akosombo hydroelectric plant facility established an FPV plant capacity factor of 14.1%.
- The total annual electricity that can be delivered by 420 MWp floating PV system is 520,233 MWh. The projected FPV system as a clean source of energy, will reduce CO₂ emissions by 308,904.5 tCO₂/MWh each year and GHG reduction revenue of USD \$216,233 at \$0.70/5 tCO₂ GHG reduction credit rate of Ghana.

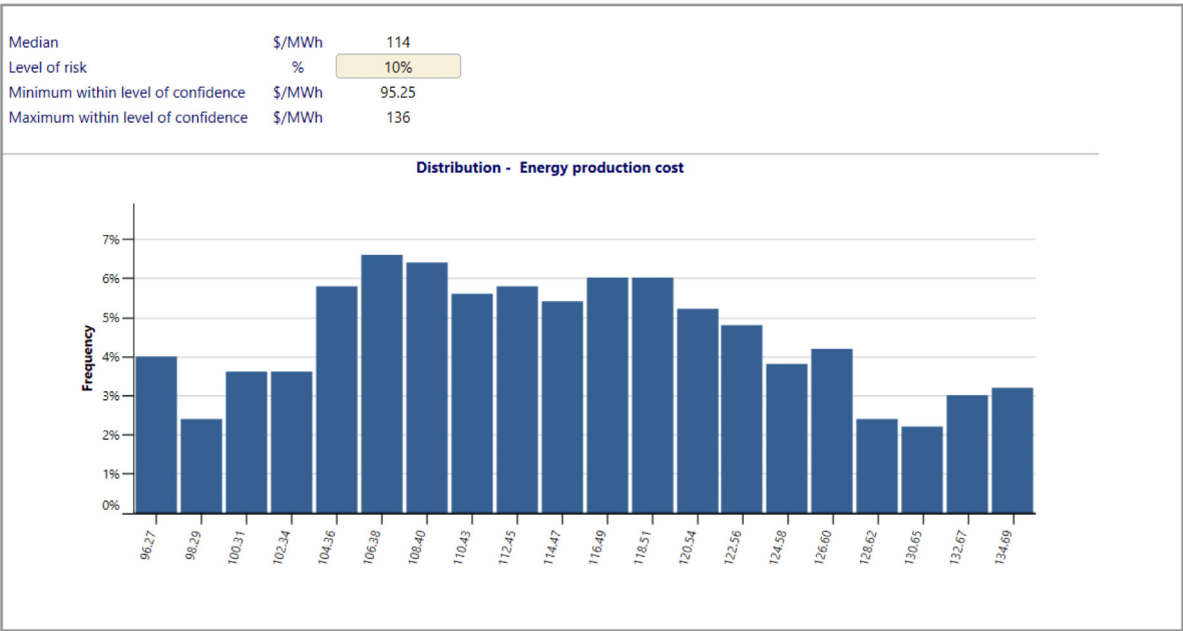


Fig. 12. FPV energy production cost distribution chart for risk analysis.

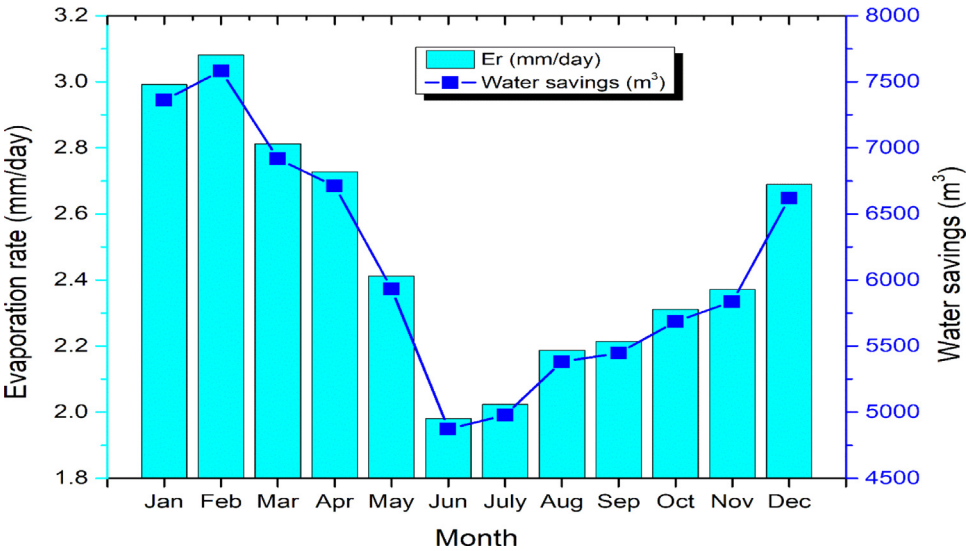


Fig. 13. Evaporation reduction and water savings potential of hybrid FPV-hydro systems.

- iii. The FPV system will save water by reducing evaporation. The projected 420 MWp system will save 73,327 m³ annually and roughly 1,833,175 m³ throughout the systems working life, covering 2,460,457 m² (2.46 km²) of Volta Lake reservoir area.
- iv. The Akosombo Dam hydroelectric plant will generate an extra 10 MWh per year when the hydro plant is coupled and run on the water savings from the FPV.
- v. For economic analysis, the normalized cost of setting up the FPV plant system is USD\$567,030,000 according to the cost analysis. At an initial capital cost of \$1,350/kW and a levelized electricity unit export cost rate of US\$ 0.10/kWh, an annual profit of USD \$52, 238,576 generated. However, the system simple payback time and debt payment term

- of 12 and 15 years respectively at a project positive net present value.
- vi. From sensitivity and risk analysis performed on the financial parameters the power system sources observed as the major cost impact relative to the other parameters. Net GHG reduction posed the least impact.
- vii. The study's finances reveal that increasing tariffs reduces the basic payback period, which is a crucial element of enhancing the hybrid plants' financial sustainability. This study suggests that a hybridized FPV-Hydropower plant is possible in Ghana and may greatly contribute to a more dependable, greener, and less expensive energy producing system, in meeting the 10% additional Renewable Energy (RE) target.

Declaration of competing interest

We the authors of the research work declare that we are in support and have no conflict of interest which had any effect on our research in any form. We therefore declare that our results are the true reflections of the outcomes as we were not faced with any conflict of interest ('Declaration of Interest: None).

Data availability

Data will be made available on request.

Acknowledgements

The authors acknowledge the efforts of all contributors, especially during the field work.

References

- Aboagye, B., Gyamfi, S., Ofosu, E.A., Djordjevic, S., 2021a. Degradation analysis of installed solar photovoltaic (PV) modules under outdoor conditions in Ghana. *Energy Rep.* 7, 6921–6931. <http://dx.doi.org/10.1016/j.egy.2021.10.046>.
- Aboagye, B., Gyamfi, S., Ofosu, E.A., Djordjevic, S., 2021b. Status of renewable energy resources for electricity supply in Ghana. *Sci. African* 11, e00660. <http://dx.doi.org/10.1016/j.sciaf.2020.e00660>.
- Ahadzie, D.K., Opoku, R., Nana, S., Ware, O., Mensah, H., 2020. Analysis of occupant behaviour in the use of air-conditioners in public buildings in developing countries: evidence from Ghana. <http://dx.doi.org/10.1108/IJBPA-01-2020-0001>.
- Anane-Fenin, K., 1986. Estimating solar radiation in Ghana. *Int. Cent. Theor. Physics, Trieste, Italy*. 1–12.
- Anon, 2023. Rets. E. D. A. S. and M. T. Natural Resources Canada Natural Resources Canada, RETScreen Data Analysis Software and Modelling Tool, n.d., [Online]. Available: <https://www.nrcan.gc.ca/energy/software-tools/7465>.
- Arjunan, K., 2019. Validity of NPV Rule and IRR Criterion for Capital Budgeting and CBA 1 Kannapiran C. Arjunan – Freelance Economic Consultant 2. SSRN, pp. 1–35. <http://dx.doi.org/10.2139/ssrn.3505058>.
- Ates, A.M., Yilmaz, O.S., Gulgen, F., 2020. Using remote sensing to calculate floating photovoltaic technical potential of a dam's surface. *Sustain. Energy Technol. Assess.* 41, <http://dx.doi.org/10.1016/j.seta.2020.100799>.
- Bui Power, A., 2019. 250MWp solar project under construction. *Annu. Rep.* 2019, 1–92.
- Chowdhury, R., Aowal, M.A., Mostafa, S.M.G., Rahman, M.A., 2020a. Floating solar photovoltaic system: An overview and their feasibility at kaptai in rangamati. In: 2020 IEEE Int. Power Renew. Energy Conf. IPRECON 2020. <http://dx.doi.org/10.1109/IPRECON49514.2020.9315200>.
- Chowdhury, N., Hossain, C.A., Longo, M., 2020b. Feasibility and cost analysis of photovoltaic-biomass hybrid energy system in off-grid areas of Bangladesh. *Sustainability* 12, 1568. <http://dx.doi.org/10.3390/su12041568>.
- Diawuo, F.A., Scott, I.J., Baptista, P.C., Silva, C.A., 2020. Assessing the costs of contributing to climate change targets in sub-Saharan Africa: The case of the Ghanaian electricity system. *Energy Sustain. Dev.* 57, 32–47.
- Drury, E., Denholm, P., Margolis, R., Drury, E., Denholm, P., Margolis, R., 2011. The Impact of Different Economic Performance Metrics on the Perceived Value of Solar Photovoltaics. NREL, Tech. Rep. NREL/TP-6A20-52197, U.S. Dep. Energy, p. 44. no. tp://www.osti.gov/bridge, doi: Task No. 5510.2210.
- Durkovic, V., Djuricic, Z., 2017. Analysis of the potential for use of floating PV power aluminium plant in Montenegro. *Energies* <http://dx.doi.org/10.3390/en10101505>.
- Energy Commission of Ghana, 2020a. Energy Commission, Ghana 2020 Energy (Supply and Demand) Outlook for Ghana. 2020, (April), pp. 1–27.
- Energy Commission of Ghana, 2020b. National energy. Natl. ENERGY Stat. 2000–2019 Strateg. 27 (9), 374–375. <http://dx.doi.org/10.1088/0031-9112/27/9/008>.
- Eshun, M.E., Amoako-Tuffour, J., 2016. A review of the trends in Ghana's power sector. *Energy. Sustain. Soc.* 6 (1), 1–9. <http://dx.doi.org/10.1186/s13705-016-0075-y>.
- Farfan, J., Breyer, C., 2018. Combining floating solar photovoltaic power plants and hydropower reservoirs: A virtual battery of great global potential. *Energy Procedia* 155, 403–411. <http://dx.doi.org/10.1016/j.egypro.2018.11.038>.
- Febrian, H.G., Supriyanto, A., Purwanto, H., 2023. Calculating the energy capacity and capacity factor of floating photovoltaic (FPV) power plant in the cirata reservoir using different types of solar panels Calculating the energy capacity and capacity factor of floating photovoltaic (FPV) power plant, <http://dx.doi.org/10.1088/1742-6596/2498/1/012007>.
- Fiabge, M., Obeng, Y.A.K., 2006. Optimum operation of hydropower system in Ghana. *J. Sci. Technol.* 26 (2), 9, Ghana.
- Ghana Grid Company Limited, 2020. 2020 Electricity Supply Plan (ESP) 2020. Ghana Grid Co. Ltd., pp. 1–58, [Online]. Available <http://www.gridcogh.com/electricitysupplyplan>.
- Global Energy Review 2020, 2020. Global Energy Review 2020. <http://dx.doi.org/10.1787/a60abbf2-en>.
- Golroodbari, S.Z.M., et al., 2021. Pooling the cable: A techno-economic feasibility study of integrating offshore floating photovoltaic solar technology within an offshore wind park. *Sol. Energy* 219 (2020), 65–74. <http://dx.doi.org/10.1016/j.solener.2020.12.062>.
- Gonzalez Sanchez, R., Kougiass, I., Moner-Girona, M., Fahl, F., Jäger-Waldau, A., 2021. Assessment of floating solar photovoltaics potential in existing hydropower reservoirs in africa. *Renew. Energy* 169, 687–699. <http://dx.doi.org/10.1016/j.renene.2021.01.041>.
- Google Earth, 2021. Google earth. https://earth.google.com/web/search/akosombo+dam/@6.2994733,0.058794,35.1293386a,1599.87959474d,35y,0h,0t,0r/data=CigijgokCW-n8vZFmhhAEXESvu3YMhRAGZR4IAmfxug_ITQM1VxAWPO_, (Accessed Jun. 18, 2021).
- Goswami, A., Kumar, P., 2021. Adoption of floating solar photovoltaics on waste water management system: a unique nexus of water - energy utilization, low - cost clean energy generation and water conservation. *Clean Technol. Environ. Policy* (0123456789), <http://dx.doi.org/10.1007/s10098-021-02077-0>.
- Gunfaus, M.T., Waisman, H., 2021. Assessing the adequacy of the global response to the Paris agreement: Toward a full appraisal of climate ambition and action. *Earth Syst. Gov.* (xxxx), 100102. <http://dx.doi.org/10.1016/j.esg.2021.100102>.
- Gyamfi, S., Modjinou, M., Djordjevic, S., 2015. Improving electricity supply security in Ghana—The potential of renewable energy. *Renew. Sustain. Energy Rev.* 43, 1035–1045. <http://dx.doi.org/10.1016/j.rser.2014.11.102>.
- Hove, A., Qian, W., Zhao, K., Fuerst, N.K., 2020. China Energy Transition Status Report 2020, pp. 1–106, [Online]. Available: https://www.energypartnership.cn/fileadmin/user_upload/china/media_elements/publications/China_Energy_Transition_Status_Report.pdf.
- Huld, T., Commission, E., 2014. Mapping the cost of electricity from grid-connected and off-grid PV systems in Africa. In: Prepr. 1st Africa PVSEC, Durban, South-Africa. vol. 2014, (September), <http://dx.doi.org/10.5071/1stAfricaPVSEC2014-3AO.2.4>.
- IEA, 2019. World Energy Outlook 2019. pp. 1–408, [Online]. Available: <https://www.iea.org/reports/world-energy-outlook-2019>0Ahttps://webstore.iea.org/download/summary/2467?fileName=japanese-Summary-WEO2019.pdf.
- IRENA, 2019. Renewable capacity statistics 2019.
- Ismail, M.A., Waqas, M., Ali, A., Muzzamil, M.M., Abid, U., Zia, T., 2022. Enhanced index for water body delineation and area calculation using google earth engine: a case study of the manchar lake. *J. Water Clim. Chang.* 13 (2), 557–573. <http://dx.doi.org/10.2166/wcc.2021.282>.
- Kim, S.M., Oh, M., Park, H.D., 2019. Analysis and prioritization of the floating photovoltaic system potential for reservoirs in Korea. *Appl. Sci.* 9 (3), <http://dx.doi.org/10.3390/app9030395>.
- Kougiass, I., Szabó, S., Monforti-Ferrario, F., Huld, T., Bódis, K., 2016. A methodology for optimization of the complementarity between small-hydropower plants and solar PV systems. *Renew. Energy* 87, 1023–1030. <http://dx.doi.org/10.1016/j.renene.2015.09.073>.
- Lee, N., et al., 2020. Hybrid floating solar photovoltaics-hydropower systems: Bene fits and global assessment of technical potential. *Renew. Energy* 162, 1415–1427. <http://dx.doi.org/10.1016/j.renene.2020.08.080>.
- Liu, L., Sun, Q., Li, H., Yin, H., Ren, X., Wennersten, R., 2019. Evaluating the benefits of integrating floating photovoltaic and pumped storage power system. *Energy Convers. Manag.* 194 (May), 173–185. <http://dx.doi.org/10.1016/j.enconman.2019.04.071>.
- Liu, L., Wang, Q., Lin, H., Li, H., Sun, Q., 2017. Power generation efficiency and prospects of floating photovoltaic systems. *Energy Procedia* 105 (2018), 1136–1142. <http://dx.doi.org/10.1016/j.egypro.2017.03.483>.
- Mittal, D., Saxena, B.K., Rao, K.V.S., 2017. Potential of floating photovoltaic system for energy generation and reduction of water evaporation at four different lakes in Rajasthan. pp. 238–243.
- Moner-girona, M., Sanchez, R.G., Kougiass, I., Moner-girona, M., Fahl, F., 2021. Assessment of floating solar photovoltaics potential in existing hydropower reservoirs in Africa. *Renew. Energy* 169 (January), 687–699. <http://dx.doi.org/10.1016/j.renene.2021.01.041>.
- Monforti-ferrario, F., Huld, T., Katalin, B., 2016. A methodology for optimization of the complementarity between small- hydropower plants and solar PV systems. *Renew. Energy* 87 (2016), 1023–1030. <http://dx.doi.org/10.1016/j.renene.2015.09.073>.
- Murat, A., Salih, O., Gulgen, F., 2020. Using remote sensing to calculate floating photovoltaic technical potential of a dam's surface. *Sustain. Energy Technol. Assess.* 41 (October), 100799. <http://dx.doi.org/10.1016/j.seta.2020.100799>.
- Oliveira-pinto, S., Stokkermans, J., 2020. Assessment of the potential of different floating solar technologies – overview and analysis of different case studies. *Energy Convers. Manag.* 211 (2019), 112747. <http://dx.doi.org/10.1016/j.enconman.2020.112747>.

- Opoku, R., Adjei, E.A., Ahadzie, D.K., Agyarko, K.A., 2020a. Energy efficiency, solar energy and cost saving opportunities in public tertiary institutions in developing countries: The case of KNUST, Ghana. *Alex. Eng. J.* 59 (1), 417–428. <http://dx.doi.org/10.1016/j.aej.2020.01.011>.
- Opoku, R., Mensah-darkwa, K., Muntaka, A.S., 2018. Techno-economic analysis of a hybrid solar PV-grid powered air-conditioner for daytime office use in hot humid climates – A case study in Kumasi city, Ghana. *Sol. Energy* 165 (2017), 65–74. <http://dx.doi.org/10.1016/j.solener.2018.03.013>.
- Opoku, R., Obeng, G.Y., Adjei, E.A., Davis, F., Akuffo, F.O., 2020b. Integrated system efficiency in reducing redundancy and promoting residential renewable energy in countries without net-metering: A case study of a SHS in Ghana. *Renew. Energy* 155, 65–78. <http://dx.doi.org/10.1016/j.renene.2020.03.099>.
- Philip, K.A., Prosper, M., Bekoe, W., 2019. Electricity supply in Ghana: The implications of climate-induced distortions in the water-energy equilibrium and system losses. *Renew. Energy* 134 (2019), 1114–1128. <http://dx.doi.org/10.1016/j.renene.2018.09.025>.
- Qi, L., Jiang, M., Lv, Y., Zhang, Z., Yan, J., 2021. Techno-economic assessment of photovoltaic power generation mounted on cooling towers. *Energy Convers. Manag.* 235 (March), 113907. <http://dx.doi.org/10.1016/j.enconman.2021.113907>.
- Rauf, H., Gull, M.S., Arshad, N., 2019. Integrating floating solar PV with hydro-electric power plant: Analysis of ghazi barotha reservoir in Pakistan. *Energy Procedia* 158, 816–821. <http://dx.doi.org/10.1016/j.egypro.2019.01.214>.
- Rauf, H., Gull, M.S., Arshad, N., 2020. Complementing hydroelectric power with floating solar PV for daytime peak electricity demand. *Renew. Energy* 162, 1227–1242. <http://dx.doi.org/10.1016/j.renene.2020.08.017>.
- Sahu, A., Yadav, N., Sudhakar, K., 2016. Floating photovoltaic power plant: A review. *Renew. Sustain. Energy Rev.* 66, 815–824. <http://dx.doi.org/10.1016/j.rser.2016.08.051>.
- Semeskandeh, S., Hojjat, M., Abardeh, M.H., 2022. Techno – economic – environmental comparison of floating photovoltaic plant with conventional solar photovoltaic plant in northern Iran, no. April. pp. 353–361.
- Sinha, S., Chandel, S.S., 2014. Review of software tools for hybrid renewable energy systems. *Renew. Sustain. Energy Rev.* 32, 192–205. <http://dx.doi.org/10.1016/j.RSER.2014.01.035>.
- Sukarso, A.P., Kim, K.N., 2020. Cooling effect on the floating solar PV: Performance and economic analysis on the case of west java.
- Teixeira, L.E., Caux, J., Beluco, A., Bertoldo, I., Louzada, J.A.S., Eifler, R.C., 2015. Feasibility study of a hydro PV hybrid system operating at a dam for water supply in southern Brazil. (September), pp. 70–83.
- Yashas, V.V., Aman, Bagrecha, 2022. Feasibility study of floating solar panels over lakes in Bengaluru City, India. 174, pp. 1–10.
- Zhao, G., Yu, B., An, R., Wu, Y., Zhao, Z., 2020. Energy system transformations and carbon emission mitigation for China to achieve global 2 °C climate target. 292, p. 2021.